

4 Laws and Constraints, Symbols and Languages

In an earlier paper, “How Does a Molecule Become a Message?” written for the 28th Symposium of the Society for Developmental Biology (Pattee 1970a), many of the ideas in my chapter “The Problem of Biological Hierarchy” (Pattee 1970b), in the third volume of *Towards a Theoretical Biology*, were further developed. I would like to add here a summary of my more recent thoughts on these fundamental problems as they have evolved since writing those papers.

In “The Problem of Biological Hierarchy” I state as a central problem of general theoretical biology, “(...) to explain the origin and operation (including the reliability and persistence) of the hierarchical constraints which harness matter to perform coherent functions.” I might have better added, the origin, operation, and *evolution* of these constraints, since new hierarchical levels evolve repeatedly. And I conclude there that in order to find an explanation at any level “(...) we will have to understand what we mean by a record or a language (...) ultimately in terms of physical concepts. We will have to learn how collections of matter produce their own internal descriptions.”

The difficulty is that any kind of description presupposes some form of language structure. As I conclude in “How Does a Molecule Become a Message?”—“A molecule becomes a message only in the context of a larger system of physical constraints which I have called a “language.” Now a language must be a closed set of structures, which we call symbols, some rules for joining and transforming these symbols which we call the grammar, as well as a set of rules for interpreting the meaning of such a collection of these symbols (for example, (Harris 1968)). What has happened is that I have begun with the problem of explaining a *constraint*, which may at first sight appear to be a relatively simple physical concept, but end up trying to explain a *language* which is a very abstract concept that no one fully comprehends. It may seem that I am trying to explain the simple in terms of the complex.

Reprinted from *Towards a Theoretical Biology* 4, C. H. Waddington, Ed. Edinburgh University Press, 1972, pp. 248–258.

I want here, in this present note, to re-emphasize my conviction that dependence on symbol structures and language constraints is the essence of life; that it is in fact the *objective* separation of the simplified symbolic description of the organism from its detailed physical reactions, that is, the separation of genotype and phenotype, which makes the evolution of living systems so profoundly different from non-living systems. Now you will say, what is new about this idea? The genotype and phenotype have been distinguished for well over a hundred years, and it is also well known that evolution depends on this separation. Yes, that much is true, but it is not enough to recognize the universal occurrence of this separation in living matter. To the physicist this separation of genetic description and phenotypic function is not a trivial question, and furthermore the meaning of these concepts does not appear clearer as we discover more details. In fact the acceptance of the structural data of molecular biology as “the physical basis of life” tends to obscure the basic question rather than illuminate it. We are taught more and more to accept the genetic instructions as nothing but ordinary macromolecules, and to forget the integrated constraints that endow what otherwise would indeed be ordinary molecules with their symbolic properties.

What I would like to counteract is the oversimplification, or perhaps what is better called the evasion of the genotype-phenotype distinction. In order to have an explanation of life, this distinction cannot be treated as merely a descriptive convenience for what is popularly assumed to be the underlying molecular basis of life. It is not the structure of molecules themselves, but the internal *self-interpretation* of their structure as symbols that is the basis of life. And this interpretation is not a property of the single molecule, which is only a symbol vehicle, but a consequence of a coherent set of constraints with which they interact.

I would like to explain why the physical concept of constraint especially in the context of origins and evolution, is by no means as simple as elementary physics textbooks lead us to believe. I would also like to suggest though I am far from implementing this suggestion, that the concept of language need not be as complex as linguists teach us. Part of this discrepancy is simply our familiarity with what are usually the most complex language structures, such as man’s spoken language, and with very simple constraints, such as a table top. But the concepts of constraint and language are very general, and closely related at a deep level. In fact, I have argued that the most fundamental concept of a constraint in physics depends on an alternative description, and that the apparent simplicity of constraints is in fact a property of the language in which it is described. On the other hand, the most elementary concept of language requires a coherent set of constraints to form its rules of grammar and interpretation.

Thus we have the chicken-egg paradox in a new form: ‘Which came first the language or the constraint?’

Laws and constraints. The concept of natural law in physics is quite distinct from the concept of a constraint. A natural law is inexorable and incorporeal, whereas a constraint can be accidental or arbitrary and must have some distinct physical embodiment in the form of a structure. Very often, however, especially

in abstract mathematical descriptions of dynamical systems, there is no *formal* way of distinguishing the laws from the constraints. Simply writing down a system's dynamical equations of motion, for example,

$$dx_i / dt = f_i(x_1 \dots x_n, t)$$

in a general system of coordinates does not reveal whether it is obeying only laws or constraints or both. If we want to predict only the local behavior of the system, this distinction may be unnecessary, but in the context of evolution the only crucial question is how the constraints themselves came about in the course of time.

Constraints, unlike laws of nature, must be the consequence of what we call some form of material structure, such as molecules, membranes, typewriters, or table tops, these structures may be static or time-dependent, but in either case it is important to realize that they are made up of matter which at all times obeys the fundamental laws of nature in addition to behaving as a constraint. What does this mean? If the laws of motion are complete and inexorable, what more can be said? Why isn't an equation of constraint either redundant or inconsistent?

The reason that constraints are not redundant or inconsistent with respect to the laws of motion is that they are *alternative descriptions* of the system. Constraints originate because of a different definition or classification of the system boundaries or system variables even though the equations of constraint may be in the same mathematical form as equations of motion. Our usual justification for choosing to use such auxiliary conditions in place of the detailed dynamics is that it *simplifies* our description of the behavior of the system. For example, the collection of molecules which make up a table top is hopelessly complex as a microscopic dynamical system, but for many practical purposes (that is, its function) it can be alternatively described as a static constraint in the form of a fixed plane, say, $z = \text{constant}$. Fixed constraints are very useful simplifications, but it is the time-dependent constraints, especially the non-integrable constraints, which generate new levels of organization.

Constraints, rules, and regulations. Time-dependent constraints often appear to us as embodiments of rules. Thus a very complex dissipative arrangement of molecules may function as a switch, in which case we often choose to ignore all the dynamical complexity and express the constraint by a simple switching rule. Complex systems may be described by many levels of constraints depending on the degree of simplification which is useful. Thus the abstract switching rule may be useful for a logical analysis of ideal networks, but much too simple for an analysis of the reliability of this function in any real computer.

It is also easy to extend the somewhat subjective concept of *rule* to its more active meaning of *regulation*, which is defined, among other things, as "control according to a rule". Control constraints or mechanisms are, of course, a very complicated and ill-defined set of structures. But in essence control implies that a system possesses *alternative* behaviors, and that owing to the particular nature of the constraint it is possible to correlate a controlling input variable or signal with a particular alternative output dynamics according to a rule.

Again it is important to realize that *controls must operate between different descriptive levels*, just as constraints must be defined by different descriptive levels. This is necessarily the case for all measurement, recording, classification, decision-making, and informational processes in which a number of alternatives on one level of description is reduced by some evaluative procedure at a higher level of description. Why are these necessarily two-level processes? Why are two distinct descriptions necessary? Because we cannot speak of an event as being both possible and impossible using the same level of description. On the lower, unconstrained level the alternatives must be possible; for if they were impossible then deciding for or against them would be a vacuous process. But on the upper, constrained or controlled level, in so far as the rules are reliable or effective, some of these alternatives are actually selected, or more precisely, made more probable, that is, catalyzed (Pattee 1972). This is one fundamental reason for the necessity of hierarchical levels of control which are characteristic of biological organization (Pattee 1970b). I suspect it is also at the root of the measurement problem in which the description of physical events (equations of motion) cannot be used directly for the description of the measurement of these same events (Pattee 1971a). This argument is also very similar to the logician's argument that any description of the truth of a proposition must be in a richer language (metalanguage) than that in which the proposition itself is stated (object language).

But now we are speaking as an outside observer who chooses his descriptions quite subjectively for the purpose of simplifying a problem he wishes to solve or for providing a simple function by a clever use of controls. Living systems on the other hand are created by their constraints and function quite independently of the biologist or physicist who studies them. How do spontaneous constraints arise in matter? Or more exactly, how do we recognize inherent constraints which have evolved autonomously, rather than from the observer's search for simplification or control? How do we distinguish the living system's rules and functions from our subjective attempts to describe such enormously complex systems?

Does life depend on laws or constraints? At this point I believe that the naive realism, characteristic of classical physics and modern biology, runs into serious difficulty when it is applied to explanations or reductions of life to physical laws. Some constraints could in principle be looked at in terms of their detailed dynamics and would be found to obey the laws of motion. This is the way we try to answer the question "How does it work?" when we are presented with a complicated machine or functional constraint. But as I point out in "The Problem of Biological Hierarchy", the question, "How did the functional constraints arise?" cannot be answered so directly because, as we have said, the constraint is not a deterministic consequence of the detailed dynamics of the system but an *alternative description* of the whole system taken as a functional unit.

Now if this alternative description is regarded simply as the outside observer's way of handling the complexity; that is, if the concepts of constraint, function, hierarchical levels, genotype and phenotype, and so on, are regarded as only a useful manner of speaking about the underlying physics and chemistry, then there is no

objective difference between living and non-living matter. From this perspective, life appears as nothing but a very complex physical system which we as observers are forced to describe in terms of hierarchical levels of organization. Consequently, in this view the origin of life and its macroscopic evolution is regarded as only a gradual increase in complexity which has necessitated new forms of description on the part of the observing biologist.

This attitude of naive realism runs against almost all modern interpretations of physics. To regard the distinction between genotype and phenotype as based on only a kind of historical biological utility, which can now be replaced more accurately by the ‘newly-discovered’ underlying physics and chemistry of nucleic acid molecules and enzymes (that is, “the physical basis of life”) is very much like claiming that symbols and records can be accurately understood by a detailed physical and chemical analysis of the symbol vehicle or record structure. We know that this is not the case, and that symbol vehicles are largely arbitrary “frozen accidents.” It is only the integrated set of rules of grammar and interpretation that gives these particular physical structures their symbolic attributes. What constitutes an “integrated set” or a “language” is of course the basic problem.

In physics, this problem arises in the concept of measurement. Measuring devices are non-integrable constraints which classify and record alternatives, and are not subject to detailed description in terms of the underlying dynamics, even in principle. Measurements are the result of the interaction of a microscopic dynamical system with a special type of dissipative constraint that can so far only be understood by its alternative, statistical description. Furthermore, physicists often regard the dissipative measurement process as more fundamental than the unobservable, formal determinism of the dynamical laws. Thus Born, for example, argues that observation itself is primary: “for whether in a concrete case a cause-effect relation holds can only be judged by applying the laws of statistics to the observations” (Born 1964). And Wigner in a recent review of the epistemology of quantum mechanics concludes: “In my opinion, the restriction of quantum mechanical theory to the determination of the statistical correlations between subsequent observations reproduces most naturally the spirit of that theory” (Wigner 1971).

In any case, whether one regards the laws or the measuring constraints as primary, it is well known among physicists (and should be better known among biologists) that a clear and unified description of events and records of events has not yet proved possible in quantum theory. It is for this reason that constraints in physics need not be so simple; and in fact it is precisely for those nonintegrable constraints which are necessarily associated with the writing and reading of symbols—whether for genetic records, controls, or measurements that unity and objectivity is lacking. Furthermore, if this unity is lacking between events and records of events in physics, then it is not easy to understand how, through the facts of molecular biology, this unification can appear.

On the other hand, by the study of *evolutionary theories*, in particular, theories of the spontaneous origin of life and its hierarchical levels of control, I think we find clues to the solution or at least to the difficulties of finding an objective basis for the separation of matter and symbols. What are some of these clues?

Self-constraint and self-rule. It is my central idea or strategy that *the essence of the matter-symbol problem and hence the measurement or recording problem must appear at the origin of life* where the separation of genotype and phenotype through language structures took place in the most elementary form. Studying the problem in this context helps remove the physicist and logician from the measurement problem. At least it helps widen his tacit anthropocentric assumptions about the function of measurement processes, shifting the emphasis from his brain to the environment or ecosystem where selection takes place. Furthermore, at the level where life originated, the problem must appear greatly reduced in complexity, although it may well be necessary to generalize our concept of records, symbols, and languages to apply them at that primeval level. The relation of this approach to the more conventional physical study of the quantum theory of measurement I have discussed in “Can Life Explain Quantum Mechanics?” (Pattee 1971a).

The necessity of an objective criterion for the occurrence of a separation of events and records has also led me to the hypothesis that *the constraints of living matter must contain their own descriptions*. This follows from the physicist’s concept of a constraint as an alternative description of an underlying dynamical process which suffers, as we said earlier, from a basic subjectivity. How do we know that a constraint is nothing more than a convenience for the higher purposes of human computation or control? The only answer I have found is in objectifying the description itself. But of course this shifts the problem to the objective description of the description which at first sight sounds like the beginning of an infinite regress. It is here that we must turn to the fundamental property of a language which, as we said at the beginning, is itself a larger coherent system of constraints. How does this help evade the infinite regress?

If one constraint gets us into this difficulty, how can adding more than one get us out?

This same problem is stated for a language in the following form: How is one symbol or word given a definite meaning? By a coherent set of other words which we call a definition. Then how are the words of the definition given meaning? This also sounds like an infinite regress, but we know that in language this problem is solved. There exist many finite sets of words which can not only define themselves, so to speak, but also define the grammar, as well as form meaningful statements about symbols or groups of symbols, which are called metalinguistic statements. A language therefore possesses the property of self-description or, in the more physical terminology, self-constraint.

There remain, of course, many fundamental questions about language. What is the simplest self-constraining set? Can simpler sets be expected to evolve this self-referent property? How do the grammars of such sets evolve? This type of question tends to emphasize the abstract, symbolic aspects of language, but what I believe more important for biology are the physical properties which make possible symbolic behavior in the first place. Even the most abstract symbols must have a physical embodiment, however arbitrary the symbol vehicle structure may be. Instead of requiring simply a finite, *self-defining* system in the abstract symbolic sense, it is more fundamental to require a finite, *self-constructing* system in the direct physical sense.

This implies a set of constraints which in some co-ordinated way can reconstruct themselves, as well as establish rules by which other structures can be generated. This coordinated set of constraints would amount to a language structure that creates a new hierarchical level of organization by allowing alternative descriptions of the underlying detailed behavior. But the problem of the spontaneous origin of such a set remains.

The origin problem. There are two very general pictures that one may form of how coordinated sets of constraints might arise spontaneously from more or less chaotic beginnings. The most common picture is that of elementary units assembling themselves into larger units. For example, the origin of life is commonly pictured as starting with simple molecules of carbon dioxide, methane, ammonia, and water, and through the activation of some energy source gradually building up amino acids, sugars, bases, then polypeptides and polynucleotides, and finally cell-like aggregations of these macromolecules, until eventually the minimum complexity of living cells is finally reached from which biological evolution can proceed. Some stages of this picture have already been demonstrated in abiogenic experiments. This sequence of origin might be called the formation of *complexity from simplicity*.

A more subtle and less easily pictured process of spontaneous organization can come about the other way around. We imagine initially chaotic aggregations of extreme complexity within which there arise persistent regularities which, so to speak, condense out simple behavior. This formation of *simplicity from complexity* is inherently a collective or global activity of the entire aggregation, as opposed to the locally specific nature of organization created by aggregation of special structures. It is this latter picture of spontaneous organization that accounts more easily for the origin of new hierarchical levels of control, which includes integrated sets of constraints that I have called language structures. For example, while specific enzyme-like polypeptides might arise by the spontaneous-assembly process of our first picture, it is unlikely that this same process could account for the genetic coding enzymes that require an integrated set of highly cooperative constraints.

Unfortunately, specific origins of the second type are much more difficult to model or demonstrate experimentally since they depend on the detailed properties of a globally complex system. On the other hand, the general type of simplification from complexity has been illustrated by models, for example, Thom's catastrophes in his topological dynamics (Thom 1970), dissipative structures in nonequilibrium thermodynamics (for example, (Prigogine et al. 1969)), and Levins' spontaneously simplifying complex systems (Levins 1970). The behavior of Kauffman's randomly connected, random switching nets also illustrates this behavior in fixed sequential systems (Kauffman 1970).

These models and theories are of great help in biology because of their generality and applicability to the problem of the internal generation of constraints. However, they lack many of the known physical and chemical interactions that in my opinion lead to the formation of the most significant constraints. For example, it is hard for me to imagine how the origin of selective, catalytic growth characteristic of present cells can be modeled without including some representation of selective monomer

addition steps in individual growing copolymer molecules. In other words, the system must generate internal constraints at the molecular level as well as at the statistical or macroscopic level.

On the other hand, all the demonstrations of abiogenic syntheses at the chemical level, while of great significance for verifying the complexity-from-simplicity theories, lack a realistically complex macroscopic environment. Since the initial conditions of these experiments are kept as simple as possible, they are unlikely to generate constraints of the simplicity-from-complexity type.

Can we test origin theories? How can we design experiments that could demonstrate the spontaneous appearance of constraints that internally simplify the behavior of a complex system? How do we recognize such simple behavior as inherently generated by objective constraints rather than by the outside observer's subjective classifications of the system's behavior?

The first requirement is that the choice of initial conditions be guided by realistic appraisal of the complexity of any primitive earth environment, and not by pre-experimental selection on the part of the designer of what is presumed to be significant. We should think of the initial problem as one of accurately simulating a primitive, sterile ecosystem, rather than biasing our analysis in favor of specific reactions or products. I have for some time suggested this type of simulation as a complement to the many abiogenic synthesis demonstrations (Pattee 1965; Pattee 1971b; Pattee 1973; Pattee 1971c). My first choice would be a simulated sterile seashore with primitive atmosphere, diurnal radiation, sand and clay, waves and tides, and so on, all of which could not have been reasonably missing from the primeval seashore, and all of which seem likely to have significant effects on the chemical development of almost all the reactants, especially any copolymers. Our initial observations of such a system must be more in the style of a naturalist rather than that of a biochemist. But as regularities or predominant behavior appear we must determine if specific catalytic, structural, or thermodynamic constraints are responsible. By their very nature, such simulations are large and expensive and will require cooperative effort from several fields for their design and monitoring. On the other hand, compared to the size and cost of modern exobiology experiments in the form of planetary explorations, such terrestrial simulations are relatively inexpensive and can be expected to provide essential data that could not be obtained any other way.

Even if such realistically complex simulations should develop internal, coordinated constraints relevant to origin problems, this is not equivalent to a theory of origins. A theory need not provide detailed predictions, but at least one might hope that from the theory it would be possible to eliminate unessential complications which may occur in primitive earth simulations. In other words, one would hope to apply the theory to computer programs that might then evolve new hierarchical levels of organization. One difficulty is that all present programming languages operate in a purely symbolic, sequential mode with no necessary physical constraints. As Conrad has suggested in this volume (Conrad 1972), the cost of achieving algorithmic prescription of sequential operations may be the loss of precisely

those global condensations that are essential for self-constraint or self-simplification. Whatever the case, we shall learn more about origins only by serious theoretical and experimental study of the problem—the same course that we follow in other sciences.

References

- Born, M. (1964). *The natural philosophy of cause and chance* (p. 47). New York: Dover Publications.
- Conrad, M. (1972). The importance of molecular hierarchy in information processing. In C. H. Waddington (Ed.), *Towards a theoretical biology* (Essays, Vol. 4, pp. 222–228). Edinburgh: Edinburgh University Press.
- Harris, Z. (1968). *Mathematical structures of language*. New York: Wiley.
- Kauffman, S. (1970). Behaviour of randomly constructed genetic nets: Binary element nets (p. 18) Continuous element nets (p. 38). In C. H. Waddington (Ed.), *Towards a theoretical biology* (Drafts, Vol. 3). Edinburgh: Edinburgh University Press.
- Levins, R. (1970). Complex systems. In C. H. Waddington (Ed.), *Towards a theoretical biology* (Drafts, Vol. 3, p. 73). Edinburgh: Edinburgh University Press.
- Pattee, H. (1965). The recognition of hereditary order in primitive chemical systems. In S. Fox (Ed.), *The origin of prebiological systems* (p. 400). New York: Academic Press.
- Pattee, H. (1970a). How does a molecule become a message? *Developmental Biology*, 3(Suppl.), 1–16.
- Pattee, H. (1970b). The problem of biological hierarchy. In C. H. Waddington (Ed.), *Towards a theoretical biology* (Drafts, Vol. 3, p. 117). Edinburgh: Edinburgh University Press.
- Pattee, H. (1971a). Can life explain quantum mechanics? In T. Bastin (Ed.), *Quantum theory and beyond* (p. 307). Cambridge: Cambridge University Press.
- Pattee, H. (1971b). The recognition of description and function in chemical reaction networks. In R. Buvet & C. Ponnampertuma (Eds.), *Chemical evolution and the origin of life*. Amsterdam: North Holland.
- Pattee, H. (1971c). Physical theories of biological coordination. *Quarterly Reviews of Biophysics*, 3, 255.
- Pattee, H. (1972). Physical problems of decision making constraints. *International Journal of Neuroscience*, 3, 99.
- Pattee, H. (1973). Physical problems of the origin of natural controls. In A. Locker (Ed.), *Biogenesis evolution and homeostasis*. Heidelberg/New York: Springer.
- Prigogine, I., Lefever, R., Goldbetter, A., & Herschkowitz-Kauffman, M. (1969). Symmetry breaking instabilities in biological systems. *Nature*, 223, 913.
- Thom, R. (1970). Topological models in biology. In C. H. Waddington (Ed.), *Towards a theoretical biology* (Drafts, Vol. 3, p. 89). Edinburgh: Edinburgh University Press.
- Wigner, E. P. (1971). Observations at the end of the conference. Preprint for *On the epistemology of quantum mechanics*. London/Ontario, March 1971.